# TED ZHAO

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## PROFILE

- Over 5 years of experience in Machine Learning and Data Science, specializing in recommendation systems and computer vision.
- Experienced in delivering end-to-end ML solutions using Databricks/Spark, Docker, and Python.
- Skilled in conducting AB tests and tracking metrics for real-world business applications.

## SKILLS

| | |
|---|---|
| **Programming Languages** | **Python, VBA, Scala, Julia** |
| **AI/Machine Learning** | Reinforcement Learning, Deep Learning, Clustering, Lightgbm, XGBoost, SVM, Word2Vec |
| **Data Science** | Tableau, A/B Testing, Statistics, Data Analysis |
| **ETL** | SQL, Spark, Databricks, Pandas |
| **Version Control** | Git |
| **Containerization** | Docker |

## WORK EXPERIENCE

### Machine Learning Engineer - Skyscanner

*Shenzhen, China (Collaborated with teams across the UK and Bulgaria)* *Nov 2019 - July 2021 (1 year and 9 months)*

- **[Recommendation Systems]** Developed and improved recommendation systems using iterative ML models (Logistic Regression / LightGBM / Deep learning).
  - Utilized Docker for end-to-end model delivery, resulting in a **+5-10% increase in CTR** with each iteration.
- **[Computer Vision / Deep Learning]** Applied Deep learning models to hotel images to enhance user experience.
  - Employed ResNet for categorizing hotel images into 20 landscape/interior labels, achieving **99% accuracy** in Top-5 predictions.
  - Utilized MobileNet to select thumbnails for hotels, resulting in a **+3% increase in CTR**.
- **[A/B Testing]** Performed AB test analysis for car-hire and hotel search pages.
  - Achieved **10X growth** in car-hire during the pandemic.
  - Built data dashboards in Tableau and Databricks for stakeholder reference.
- **[ETL]** Designed and implemented data pipelines in Spark/Databricks.
  - Generated **a dozen key metrics and features** to train models and conduct AB testing analytics.

### Data Scientist - Smart Decision Technology Ltd

*Shenzhen, China* *Aug 2015 - Mar 2018 (2 years and 8 months)*

- **[Simulation]** Developed and delivered a simulation system in Python for S.F Express.
  - Utilized **SimPy** to recreate the conveying system in the Distribution Center.
  - Built a front-end page to facilitate input of site designs and parameters by business personnel.
  - Established a back-end service to store simulation parameters and results for each iteration.
  - This solution outperformed three competitors in speed and customizability, securing a **2-year contract** with S.F.Express.
- **[Credit Risk Modeling]** Designed credit scorecard workflows using Python to replace SAS.
  - Created **feature selection, binning, and WOE calculation** in Python.
  - Converted the logistics model into scorecards in Python.
- **[Segment Analysis / NLP]** Delivered client segment reports for Hong Kong Jockey Club.
  - Created segment analysis using clustering and generated data diagrams and charts in reports.
  - Built **sentiment analysis using Word2Vec** on mixed Cantonese and English restaurant reviews.

### Data Scientist - DJI

*Shenzhen, China* *April 2018 - Aug 2018 (5 months)*

- **[Competitive Analysis]** Provided support for a comprehensive campaign through competitive analysis.
  - **[Data Scraping]** Scraped retail location data from competitors.
- **[Data Presentation]** Analyzed collected dataset to identify patterns and insights for the campaign team.
  - Translated findings into actionable recommendations.
  - Empowered stakeholders to make informed decisions and optimize the product launch strategy.

## RESEARCH PROJECT

### Quantitative Trading with Reinforcement Learning - The University of Adelaide

*Adelaide, SA, Australia* *Sep 2022 - Present*

- **[Reinforcement Learning]** Conducting a research project while pursuing a master's degree in Machine Learning at The University of Adelaide.
  - Developed a trading strategy for the Crypto market using **Reinforcement Learning**.
  - Designed a reward system with Sharp Ratio, where actions were the next sell/buy quantity of all assets.
  - Created **two Deep Learning modules** to capture price signals and market sentiment.
  - **[Back Testing]** Developed a back-test with metrics **Sharp Ratio, Maximum Drawdown(MDD)** and presented profit curves with plots, considering fees and compound returns.

## Education

| | Institution | Duration |
|---|---|---|
| Master of Machine Learning | The University of Adelaide | June 2021 - June 2023 |
| Bachelor's Degree in Mathematical Sciences | Shenzhen University | September 2010 - July 2014 |

## Certifications

| | Date | Verification |
|---|---|---|
| Machine Learning - Coursera | October 2016 | Verify |
| Machine Learning Nanodegree Program - Udacity | December 2018 | Verify |

## Language Proficiency

| | Proficiency |
|---|---:|
| English | IELTS - CEFR C1 (Advanced) |
| Mandarin (Chinese) | Native |
| Cantonese (Chinese) | Native |